

SOFT COMPUTING · CSA301

Network Routing Optimization Using a Genetic Algorithm

Balancing distance, delay and congestion in a multi-metric routing problem

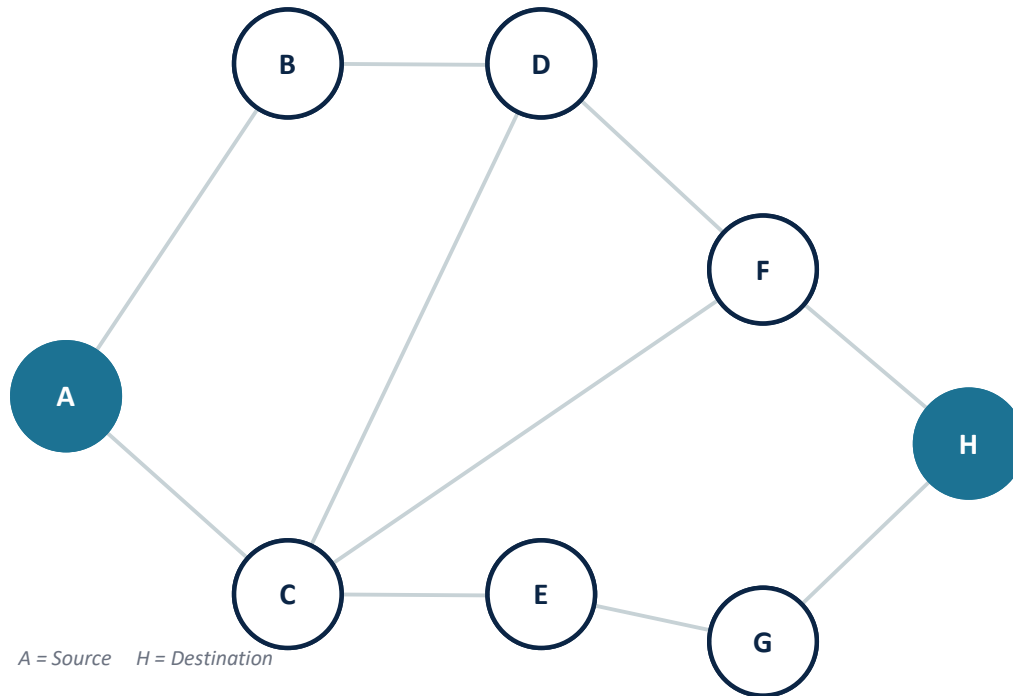
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THE PROBLEM

Is the Shortest Route Always the Best Route?

We model a communication network as a graph — and ask whether distance alone is enough to route through it.



“If one route is shorter but heavily congested, should a routing algorithm always choose it?”

Each edge carries three metrics:

- 1 Distance**
km travelled along the link
- 2 Delay**
transmission delay in ms
- 3 Congestion**
normalized load, 0 (free) – 1 (jammed)

MOTIVATION

Why Multi-Metric Routing?

A distance-only route and our multi-metric route, compared on the same network.

DISTANCE-ONLY

A → C → F → H

A

C

F

H

Distance	25 km
Delay	51 ms
Congestion	1.30
Objective cost	1.755897

MULTI-METRIC (OURS)

A → C → E → G → H

A

C

E

G

H

Distance	26 km
Delay	53 ms
Congestion	0.48
Objective cost	1.572889

Q1 · Fuzzy Logic in Real-World Routing

Vehicle / Network Information

Distance · Direction · Speed · Future direction

Fuzzy Controllers

Assess candidate relay vehicles

Route Cost + Route Lifetime

Fuzzy-evaluated route quality

Route Selection

Best relay / path chosen

Fahad, T. O. & Ali, A. A. (2018)

“Multiobjective Optimized Routing Protocol for VANETs”

Advances in Fuzzy Systems · DOI: 10.1155/2018/7210253

Uses fuzzy controllers (with ABC optimization) to combine distance, direction, speed and future direction when selecting relay vehicles in VANETs.

[View Paper →](#)

Q2 · Research Basis for Our Approach

1

Bhardwaj & El-Ocla (2020)

“Multipath Routing Protocol Using Genetic Algorithm in Mobile Ad Hoc Networks”

IEEE Access, 8, 177534–177548 · DOI: 10.1109/ACCESS.2020.3027043

How it supports our work

Applies GA to routing while considering route length, residual energy and traffic/data loss — showing GA works when several network conditions matter, not just shortest distance.

[IEEE Xplore →](#)

2

Li et al. (2013)

“A Genetic Algorithm for Finding a Path Subject to Two Constraints”

Applied Soft Computing, 13(2), 891–898 · DOI: 10.1016/j.asoc.2012.10.018

How it supports our work

Details chromosome representation, population initialization, mutation, and avoiding invalid/looping paths — directly informing how we represent and validate routes as chromosomes.

[ScienceDirect →](#)

OUR TECHNIQUE

Genetic Algorithm — Core Concepts

Each GA concept, mapped directly onto our routing problem.

1

Chromosome

One candidate route, e.g. [A, C, E, G, H]

2

Population

A collection of candidate routes

3

Fitness / Cost

Lower objective cost = better route

4

Selection

Tournament selection between candidates

5

Crossover

Two parents combined at a shared node

6

Mutation

Part of a route changed for diversity

7

Elitism

Best route always kept for next generation

These 7 mechanisms drive every generation of the search.

MATHEMATICAL FORMULATION

The Objective Function

$$\text{Cost} = 0.40 \cdot D_{\text{norm}} + 0.30 \cdot \text{Delay}_{\text{norm}} + 0.30 \cdot \text{Congestion} + 0.01 \cdot \text{Hops}$$

Lower cost = better route. Weights are our experimental design choice, not universal optimal values.

Metric	Weight
Distance	0.40
Delay	0.30
Congestion	0.30
Hop penalty	0.01 / hop

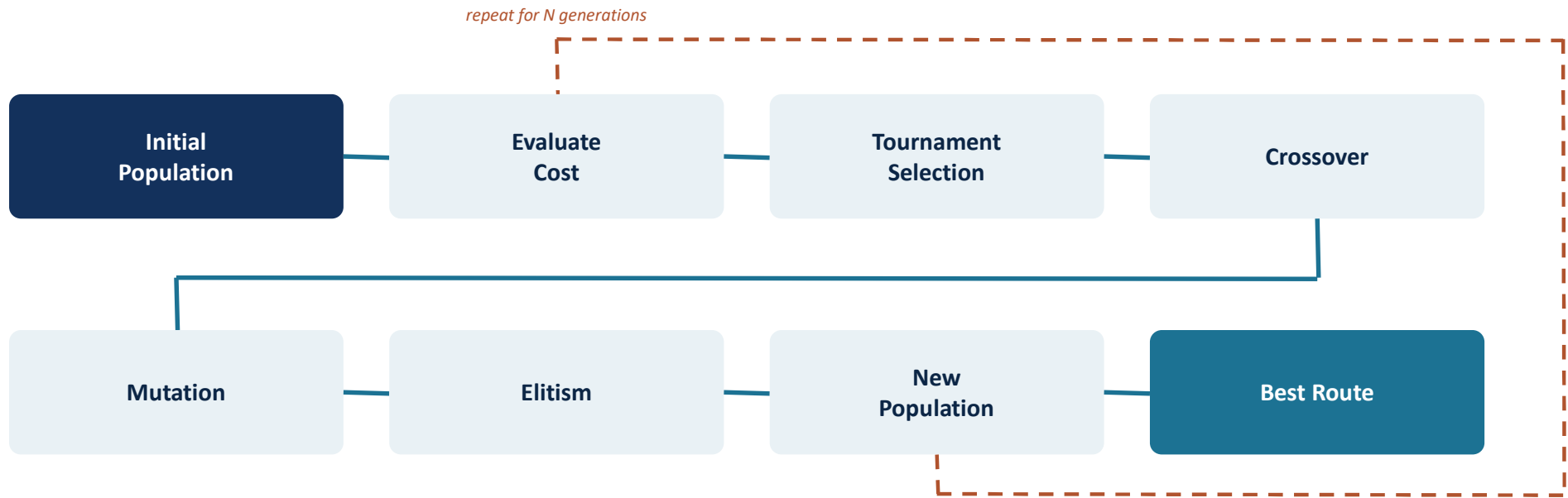
Normalization (final route: A→C→E→G→H)

26 km	→ 2.000 norm	Distance ÷ max 13 km
53 ms	→ 1.963 norm	Delay ÷ max 27 ms
0.48	already 0–1	Congestion

Final Objective Cost

1.572889

How the Genetic Algorithm Runs



Elitism guarantees the best route found so far always survives into the next generation.

RESULTS

Final GA Result



26 km

Distance

53 ms

Delay

0.48

Congestion

1.572889

Objective Cost

10 / 10

independent random seeds converged to the same route

31.73 ms

mean runtime across all seed runs

VALIDATION

Baseline Comparison

Method	Route	Distance	Delay	Congestion	Cost
Distance-only Dijkstra	A→C→F→H	25 km	51 ms	1.30	1.755897
Weighted Dijkstra	A→C→E→G→H	26 km	53 ms	0.48	1.572889
Genetic Algorithm	A→C→E→G→H	26 km	53 ms	0.48	1.572889

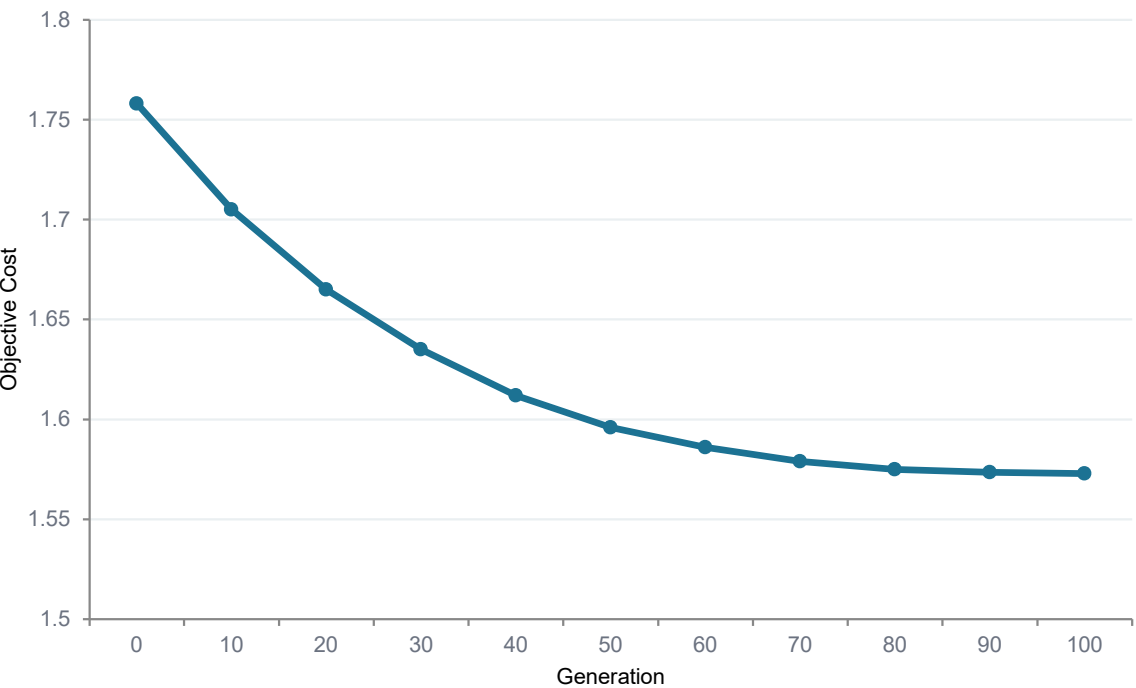
Key takeaway

The GA matches the weighted Dijkstra solution on our static additive objective — this validates our implementation, rather than showing that GA outperforms Dijkstra.

ANALYSIS

Convergence & Parameter Selection

Best cost vs. generation (representative run, seed 66)



1.758094 → 1.572889 (10.53% improvement)

Selected Parameters

Population size **16**

reached the best solution without extra computation

Mutation rate **0.10**

balances diversity against disrupting good routes

Generations **100**

sufficient headroom for the cost to stabilize

Limitations & Future Scope

Current Limitations

- Small simulated network
- Static network conditions
- Manually selected weights
- GA is inherently stochastic
- Weighted Dijkstra solves this exact static formulation

Future Work

- Dynamic congestion modeling
- Real-world network datasets
- Real-time rerouting
- Pareto multi-objective GA
- Fuzzy + GA hybrid approach
- GA combined with local search
- Larger, more complex topologies

Conclusion

- GA successfully models network routing as an evolutionary optimization problem.
- Multi-metric routing captures trade-offs that distance-only routing ignores.
- GA consistently reached the same minimum-cost route as weighted Dijkstra in our tested static network.
- The framework can be extended to dynamic and more complex routing scenarios.

REFERENCES

Fahad & Ali (2018) — Multiobjective Optimized Routing Protocol for VANETs, Advances in Fuzzy Systems. [View →](#)

Bhardwaj & El-Ocla (2020) — Multipath Routing Protocol Using GA in MANETs, IEEE Access, 8. [View →](#)

Li et al. (2013) — A Genetic Algorithm for Finding a Path Subject to Two Constraints, Applied Soft Computing, 13(2). [View →](#)